BundleFuse

A Theory of Completing Bundles

Abstract

How can we suggest an item that fits a partly completed collection? This chapter develops an answer from first principles. It begins with past collections and the items they contain, then builds several kinds of evidence: shared history, support from groups of known items, item descriptions, and categories. It explains how those pieces are put on a common scale, combined, and adjusted when an item has little history. A small numerical example follows the calculation through to a ranked recommendation.

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1 The big picture: from a partial bundle to suggestions

A *bundle* is a collection of items that belong together. A bundle could be an outfit, a shopping basket, or a music playlist. Suppose we are shown a few items from a bundle and asked to suggest the missing items. The items already shown are called the *known items*. A possible missing item is called a *candidate*.

A good candidate should fit the known items as a group. History helps: items previously found in the same bundles often fit together. Descriptions and categories help too: a new item may have no purchase or playlist history but may still resemble useful items. These clues need not agree. BundleFuse assigns a score to every candidate, combines the clues, and returns the highest-scoring items that are not already known.

1.1 One question, five steps

Imagine a store with five items, numbered 0 to 4. Someone has chosen items 0 and 1. We want to decide whether to suggest item 2, 3, or 4. Suppose two earlier baskets were $\{0, 1, 2\}$ and $\{0, 3\}$. Item 2 appeared with both chosen items, item 3 appeared with one of them, and item 4 has no basket history. The answer is not automatic: item 4 may still have a useful description, while an item seen often may simply be popular.

BundleFuse addresses this decision in five connected steps:

1. **Gather evidence.** Look at past bundles and, when available, item descriptions, categories, and other relationships. Each kind of evidence gives each candidate a score.
2. **Make the scores comparable.** A count and a description similarity have different units. Rescale each kind of evidence across the available items.
3. **Combine the scores.** Give each kind of evidence an importance, and allow extra credit when several known items support a candidate together.
4. **Account for limited history.** Blend the combined score with an estimate based on descriptions and categories, giving that estimate more influence for an item with little history.
5. **Rank and check.** Remove the items already chosen, rank the remaining items, and assess the suggestions on examples whose hidden items are known.

The following sections explain each step in this order. We start by putting past bundles into a table, because every score based on shared history needs that table. We return to the five-item example throughout, and finish its numerical calculation in Section 11. The equations make the rules precise; the paragraphs around them explain what each rule contributes to the decision.

1.2 The symbols we will use

Symbol	Meaning
$\mathcal{I}$	Set of all available items; $n = \lvert\mathcal{I}\rvert$ is the number of items.
b_e	Past bundle number e; there are m past bundles.
S	Set of known items for the bundle we want to complete.
i, s, t	Item labels: i is usually a candidate; s, t are known items.
B	A table with one row per past bundle and one column per item.
f_i	Number of past bundles containing item i.
d_e	Number of items in past bundle e.
$r_i^h(S)$	Raw evidence for candidate i from source h, given S.
$z_i^h(S)$	The same evidence after it has been put on a common scale.
w_h	Nonnegative importance assigned to source h.
$Y_i(S)$	Combined evidence from the available sources.
$P_i(S)$	A second, content-oriented estimate used when history is limited.
$\widehat{\theta}_i(S)$	Final score for candidate i.
K	Number of suggestions requested.

We use $[x]_+ = \max(x, 0)$, so $[-2]_+ = 0$ and $[3]_+ = 3$. A small positive number ε in a denominator prevents division by zero. When a number is stated as a usual setting, it is a choice of the method rather than a mathematical necessity.

2 Representing past bundles

Define the entry in row e and column i of B as

$$B_{ei} = \begin{cases} 1, & i \in b_e, \\ 0, & i \notin b_e. \end{cases} \tag{2.1}$$

The number of past bundles containing item i, and the size of bundle e, are

$$f_i = \sum_{e=1}^{m} B_{ei}, \qquad d_e = \sum_{i \in \mathcal{I}} B_{ei}. \tag{2.2}$$

For example, if the past bundles are {tea, milk, sugar} and {tea, biscuits}, then tea occurs twice, and the first bundle contains three items. If two different past bundles contain exactly the same items, they count as two observations.

For the current request, let $x_i = 1$ when $i \in S$ and $x_i = 0$ otherwise. The vector x therefore marks the known items. For our example, the table has two rows: the first marks items 0, 1, and 2, and the second marks 0 and 3. The indicator marks items 0 and 1. This gives us the basic information needed for the first question: how much does each past bundle support each candidate?

3 Evidence from shared history

3.1 Simple counts

For distinct items i and j, let C_{ij} be the number of past bundles containing both. Set $C_{ii} = 0$. In matrix form,

$$C = \operatorname{off}(B^\top B), \tag{3.1}$$

where $B^\top$ turns rows into columns and off replaces diagonal entries by zero. A simple score for i is $\sum_{s\in S} C_{is}$: add the number of times i appeared with each known item. In the example, item 2 appeared with both known items, while item 3 appeared with item 0 only. This is easy to understand, but very common items can receive large scores merely because they appear in many bundles.

3.2 Giving each past bundle a suitable weight

Let $\alpha_d \geq 0$ and $\gamma_v \geq 0$ control two adjustments. A usual choice is $\alpha_d = \gamma_v = \frac{1}{2}$. Define the weight of a known item s and the weight of a past bundle e by

$$a_s = (f_s + 5)^{-\alpha_d}, \qquad v_e = (d_e + 1)^{-\gamma_v}. \tag{3.2}$$

The first reduces the influence of a very common known item. The second reduces the influence of a very large past bundle. The constants 5 and 1 keep the weights finite and soften the adjustment for small counts.

The weighted overlap between past bundle e and the known items is

$$o_e(S) = v_e \sum_{s\in S} B_{es} a_s. \tag{3.3}$$

The first history-based score of candidate i is the sum of these overlaps over past bundles containing it:

$$p_i(S) = \sum_{e=1}^{m} B_{ei}\, o_e(S). \tag{3.4}$$

A candidate is supported when it occurs in past bundles that also contain known items. Equation (3.4) adds the contributions of known items; it does not demand that every known item be present in the same past bundle.

3.3 Extra support from several known items

Let $k_e(S) = \sum_{s\in S} B_{es}$ be the number of known items in past bundle e. Let $\beta \geq 0$ control the extra credit for group overlap; a usual value is 1.5. Define

$$\delta_i(S) = \sum_{e=1}^{m} B_{ei}\, o_e(S)\, \beta[k_e(S) - 1]_+. \tag{3.5}$$

If a past bundle contains one known item, its extra contribution is zero. If it contains two, it receives extra credit; three receive more. The strengthened history score is $p_i(S) + \delta_i(S)$. With only one known item, $\delta_i(S) = 0$ for every candidate. In our example, this extra support goes to item 2 through the first basket. The second basket cannot contribute extra group support to item 3, because it contains only one known item. We now have a direct score and a group-strengthened score. The next step asks whether connections through other items can provide additional evidence.

3.4 Following connections one step farther

A candidate can be related to the known items through other items even if the direct connection is weak. To describe this, let D be the diagonal table with entries $D_{ii} = (f_i + 5)^{-\alpha_d}$, and let E be the diagonal table with entries $E_{ee} = (d_e + 1)^{-1}$. Define the item-to-item table

$$L = \mathrm{off}(DB^\top EBD). \tag{3.6}$$

For any vector a of item scores, La spreads those scores across items that have appeared in the same past bundles, while the zero diagonal prevents an item from sending its score straight back to itself. The two longer-range scores are Lp and $L(p+\delta)$. For shopping data with additional user and session histories, a different table can take the place of L; Section 6 explains that variation.

4 When pairs occur together more often than expected

The preceding scores use observed overlaps, adjusted for the frequency of known items and the size of past bundles. There is another question: did two items occur together more often than we would expect from how common they are individually? Answering it gives us a second view of the same history.

A raw shared-history count can be large simply because both items are common. Let $g_i = \sum_j C_{ij} + \varepsilon$ be the total shared-history count of item i, and let $Z_C = \sum_{i,j} C_{ij}$ be the total of the table. For a pair observed at least once, define

$$M_{ij} = \left[\log\left(\frac{C_{ij} Z_C}{g_i g_j} + 10^{-12}\right)\right]_+ . \tag{4.1}$$

For a pair never observed together, set $M_{ij} = 0$. This number is large when the pair's count is strong relative to its overall counts, and zero when the expression is not positive. A candidate's score from this source is $\sum_{s \in S} M_{is}$.

A second source makes a shorter description of the rows of M. Choose a dimension r; find a rank-r approximation to M by a method called *singular value decomposition*, which finds directions that summarize much of a table's variation. Each item then has a short vector $u_i \in \mathbb{R}^r$; normalize each nonzero vector to length approximately one. The score is the dot product of the candidate vector with the mean known-item vector:

$$r_i^{\text{history}}(S) = u_i \cdot \left(\frac{1}{|S|} \sum_{s \in S} u_s\right). \tag{4.2}$$

A dot product adds the products of corresponding vector entries. Here it measures how closely the candidate's summary of history agrees with the average known item.

5 Evidence from descriptions and categories

History can tell us little about an item that rarely appears, and nothing about an item with no appearances. In our example, item 4 would receive no direct shared-history support. Its description or category may still tell us something useful. We now develop those sources, which will later give such items a way to compete with well-observed candidates.

5.1 Comparing item descriptions

Suppose item i has a vector c_i describing its text, image, or audio. These vectors can come from measurements or from an existing representation of the item. Normalize the vectors and compare a candidate with the mean description of the known items:

$$r_i^{\text{description}}(S) = c_i \cdot \left(\frac{1}{|S|} \sum_{s \in S} c_s\right). \tag{5.1}$$

A dot product near one means that normalized vectors point in similar directions; near zero indicates little agreement in that representation. Different types of description can produce separate scores and receive different weights.

For text, one possible description counts words and two-word phrases, gives less weight to terms appearing in many item descriptions, and then shortens the resulting table by singular value decomposition. The resulting coordinates summarize descriptions; they do not require that a candidate has appeared in a past bundle.

5.2 Separating category from the rest of a description

Items in the same category may share broad features that say little about which particular items fit together. Let c_i now denote a vector from one description type, and let $g(i)$ be the group of item i. Write $\bar{c}_g$ for the average vector of all items in group g. The remaining description after removing the group average is

$$d_i = \frac{c_i - \bar{c}_{g(i)}}{\|c_i - \bar{c}_{g(i)}\|_2 + \varepsilon}. \tag{5.2}$$

Here $\|\cdot\|_2$ is ordinary vector length. For an outfit, the group might be a clothing category; for audio, it may be a group of items with similar sound. The vectors d_i describe how items differ within their broad groups. They can be scored by the same mean-and-dot-product rule as equation (4.2).

5.3 Joining descriptions with history

Let D_c be the table whose row i is d_i. Apply L from equation (3.6) to its columns, and normalize each resulting row; call that normalized table H. For a chosen weight $\lambda \geq 0$, form and row-normalize

$$F = \text{normalize rows}(D_c + \lambda H). \tag{5.3}$$

A row of F blends an item's own description with descriptions of historically connected items. Comparing its rows using equation (4.2) supplies another score. A usual value is $\lambda = 0.5$.

5.4 Do categories complement each other?

Let $g(i)$ be the category of item i. Define A_{ec} as the number of items of category c in past bundle e, and let $a_c = \sum_e A_{ec} + \varepsilon$. Define $H = \text{off}(A^\top A)$ and $A_+ = \sum_c a_c$. For observed category pairs,

$$Q_{cd} = \left[\log\left(\frac{H_{cd} A_+}{a_c a_d} + 10^{-12}\right)\right]_+, \tag{5.4}$$

and set unobserved entries to zero. The category score for candidate i is

$$r_i^{\text{category}}(S) = \sum_{s \in S} Q_{g(s),g(i)}. \tag{5.5}$$

For example, if shirts and trousers frequently occur together, that relationship can help score trousers when a shirt is known. In some applications, an additional penalty discourages repeating a category that seldom appears twice in the same past bundle. In others, shared artist or album membership adds a separate score. The appropriate category evidence depends on what a bundle represents.

6 Additional relationships in shopping data

The past-bundle table and item descriptions are not always the only available evidence. In shopping data, for example, several baskets may belong to the same household. These relationships supply another view of how items are connected. They are optional: the reasoning up to this point works without them.

Some data contain households, users, sessions, or an additional record of user–item interactions. These can provide extra evidence without changing the purpose of the method. For example, create a table Z whose rows are households and whose columns are items, with each entry counting appearances in a household's past training baskets. Divide each household row by the square root of one plus its total count. The product of the resulting table's transpose and itself measures how often items are associated through households; its diagonal is removed. A candidate can then receive evidence when it shares household history with the known items.

For some commerce data, separate user and session tables give item-pair tables C_u and C_s, and an additional interaction record gives a weighted pair table C_p. Define $D_c = \operatorname{diag}((f_i + 1)^{-1/2})$. One longer-range table is

$$L_c = D_c(C + 0.5C_s + 0.5C_u + 0.2C_p)D_c. \tag{6.1}$$

It spreads evidence through several kinds of shared history. The particular coefficients are choices that define this version of the method. Their presence means that a result for this setting should be understood in the light of the extra information it uses.

7 Agreement among known items

So far, we have considered whether each candidate has historical, descriptive, or category support. We can also ask how that support is distributed across the known items. Is it coming from just one known item, or do several known items agree? This question matters in our example because item 2 is connected to both 0 and 1.

A candidate is especially interesting when several known items support it together. To measure this, let $q_{is} \geq 0$ be the support that known item s gives candidate i. For a concrete definition, let $W_{is} = 0$ when $i = s$; otherwise let $W_{is} = \sum_e B_{ei} v_e B_{es} a_s$, using the weights defined above. Let d_i be a normalized description vector and $\lambda_d \geq 0$ a chosen description weight. Then

$$q_{is} = W_{is} + \lambda_d [d_i \cdot d_s]_+, \qquad A_i = \sum_{s \in S} q_{is}. \tag{7.1}$$

If description vectors are unavailable, the second term is omitted.

Known items may be nearly duplicates of one another. Let G_{st} measure similarity between known items s and t. When both category labels $g(\cdot)$ and description vectors are available, one choice is

$$G_{st} = \tfrac{1}{2}\mathbf{1}[g(s) = g(t)] + \tfrac{1}{2}[d_s \cdot d_t]_+. \tag{7.2}$$

With only one kind of information, use that part alone. For distinct known items, choose a coefficient η and define

$$K_{st} = \begin{cases} [1 + \eta G_{st}]_+, & \eta < 0, \\ 1 + \eta G_{st}, & \eta \geq 0, \end{cases} \qquad K_{ss} = 0. \tag{7.3}$$

A negative η reduces the extra credit for similar known items; a positive one can increase it. Let $\tau > 0$ be a small stabilizing number, commonly 0.05. Define the agreement score

$$r_i^{\text{agreement}}(S) = \frac{\sum_{s \in S} \sum_{t \in S} q_{is} K_{st} q_{it}}{A_i + \tau}. \tag{7.4}$$

Both orders of a distinct pair are included in the double sum. If fewer than two known items are given, the score is zero. This score rewards support from pairs of known items; it does not require the candidate to match every item simultaneously.

8 Putting different evidence on the same scale

We now have several possible scores for the same candidate. To make one decision, we need to combine them. Before doing so, we must address a simple problem: counts, dot products, and category scores have different units. A count of 10 is not automatically stronger than a similarity of 0.8.

Counts, dot products, and category scores have different units. For each evidence source h and a fixed set of known items S, let $r_i^h(S)$ be its raw score for item i. Its mean across all n items is

$$\mu_h(S) = \frac{1}{n} \sum_{i \in \mathcal{I}} r_i^h(S). \tag{8.1}$$

Let $\sigma_h(S)$ be the corresponding standard deviation: a measure of how much the scores vary around their mean. The common-scale score is

$$z_i^h(S) = \frac{r_i^h(S) - \mu_h(S)}{\sigma_h(S) + \varepsilon}. \tag{8.2}$$

A positive value means above-average evidence for that source; a negative value means below average. If all items receive the same raw score, the common-scale scores are zero. The mean and spread are calculated for each request across the whole item set, before the known items are removed from the final list. Some settings use a sample standard deviation (dividing the sum of squared deviations by $n - 1$); others use a population standard deviation (dividing by n). Keeping this choice fixed matters for exact comparisons.

The strengthened history score and the simple history score are each standardized separately. Therefore, the standardized value of $p + \delta$ is generally not the sum of the separately standardized values of p and δ.

9 Combining the evidence

The preceding step expresses each kind of evidence relative to its own average and spread. We can now decide how much importance to give each kind and combine them into one score. This yields an answer based on the available observations; the next section will adjust that answer for items whose history is scarce.

Let $\mathcal{H}$ be the set of evidence sources under consideration, and let $w_h \geq 0$ be the importance of source h. For each source, let $z_i^{h,0}$ be its score with the extra group contribution removed; for sources without such a contribution, let $z_i^{h,0} = z_i^h$. Define

$$Y_i^0 = \sum_{h \in \mathcal{H}} w_h z_i^{h,0}, \qquad Y_i^1 = \sum_{h \in \mathcal{H}} w_h z_i^h. \tag{9.1}$$

Choose $a \in [0, 1]$ for the amount of group evidence, and $b \geq 0$ for the amount of agreement evidence from equation (7.4). The combined observed score is

$$Y_i = (1 - a)Y_i^0 + aY_i^1 + bz_i^{\text{agreement}}. \tag{9.2}$$

When $a = 0$, the calculation uses the simpler history scores. When $a = 1$, it uses their strengthened counterparts. The agreement term is an additional piece of evidence.

The weights are selected using held-out examples for which the hidden items are known. One can try a short list of nonnegative values for one weight at a time, retain a change when it improves a chosen ranking measure, and repeat. A small tolerance can remove a source that adds little. The amounts a and b are then chosen from small lists. This is a search over settings informed by examples; it does not require a neural network. The held-out examples used for selecting settings are kept separate from the final test examples used to report performance.

10 Making a sensible choice when history is limited

The combined score answers what the available evidence suggests. But an item seen once and an item seen hundreds of times do not have equally rich histories. In our example, item 4 has no past-bundle history at all. We therefore bring in a second estimate that can rely on information available even for that item.

A rare item has less historical evidence than a frequent one. BundleFuse therefore also forms a second estimate from selected sources such as descriptions and categories. Let P_i be their weighted, common-scale sum. Depending on the setting, it can also include the same agreement or popularity contributions used in the combined score. Let $s_0 \geq 0$ be a scale chosen using the held-out selection examples, and define the second estimate $\mu_i = s_0 P_i$.

Let $\kappa \geq 0$ control how strongly the method relies on this second estimate. If $\kappa = 0$, the final score is simply Y_i. For $\kappa > 0$, define

$$u_i = \frac{\kappa}{f_i + \kappa}, \qquad \widehat{\theta}_i = (1 - u_i)Y_i + u_i\mu_i. \tag{10.1}$$

The weights in this average add to one. An item with $f_i = 0$ receives the second estimate entirely. As f_i increases, u_i decreases, and the observed score Y_i has more influence. For example, with $\kappa = 2$, an item seen in two past bundles has $u_i = 1/2$; an item seen in 18 has $u_i = 1/10$. If the method selects $\kappa = 0$, this adjustment is disabled for every item.

10.1 A probability interpretation

The weighted average above has a useful interpretation, provided we state its assumptions. Suppose an unobserved true suitability θ_i follows a normal distribution with mean μ_i and variance σ_0^2. Suppose also that the observed score Y_i, conditional on θ_i, has mean θ_i and variance σ_e^2/f_i for $f_i > 0$. Assume σ_0^2 and σ_e^2 are positive. Then the conditional mean of θ_i given Y_i is equation (10.1) with

$$\kappa = \frac{\sigma_e^2}{\sigma_0^2}. \tag{10.2}$$

The assumption says that more appearances reduce uncertainty in the observed evidence. At $f_i = 0$, the formula uses the prior-only limit. In practice, the final κ can be chosen from several values using held-out examples, so it need not equal a single fitted variance ratio. The final scores are ranking scores, not probabilities that items belong in the bundle; the assumptions do not by themselves establish that they are calibrated.

10.2 Estimating a useful range for the adjustment

Let there be v held-out selection requests, and write Y_{ij} and μ_{ij} for item i on request j. For each item with $f_i > 0$, calculate its average squared difference between the two estimates and its inverse history count:

$$e_i = \frac{1}{v}\sum_{j=1}^{v}(Y_{ij} - \mu_{ij})^2, \qquad x_i = \frac{1}{f_i}. \tag{10.3}$$

Fit a straight line to points (x_i, e_i) across items. Its intercept suggests the prior variance and its slope suggests the extra observation variance. If there are fewer than two items with history, or all positive history counts are effectively equal, the two parts cannot be distinguished from these points and the adjustment is disabled. Otherwise, small positive lower bounds keep the estimated variances usable, and their ratio supplies one candidate value of κ. Several fixed values and this fitted value can be compared on the held-out selection examples. These squared differences compare two score estimates; they are not squared prediction errors against the hidden-item labels.

11 Putting the pieces together: the five-item example

We have now defined all five steps in the map at the beginning. Let us return to the same five items and follow a candidate from past baskets to final score. This makes clear where the group evidence and the adjustment for limited history enter.

Consider five available items numbered 0 through 4. The two past bundles are $b_1 = \{0, 1, 2\}$ and $b_2 = \{0, 3\}$; the known items are $S = \{0, 1\}$. Item 4 has no past bundle history. The history counts are $(f_0, f_1, f_2, f_3, f_4) = (2, 1, 1, 1, 0)$.

Use $\alpha_d = \gamma_v = \frac{1}{2}$ and $\beta = 1.5$. Define $a = 1/\sqrt{7}$ and $b = 1/\sqrt{6}$. The two past-bundle overlaps from equation (3.3) are

$$o_1 = \frac{a + b}{2}, \qquad o_2 = \frac{a}{\sqrt{3}}. \tag{11.1}$$

The first bundle contains both known items; the second contains just item 0. Equations (3.4) and (3.5) give, in item order,

$$p = (o_1 + o_2, o_1, o_1, o_2, 0), \quad \delta = (1.5o_1, 1.5o_1, 1.5o_1, 0, 0). \tag{11.2}$$

Item 2 receives extra group support because it appeared with both 0 and 1. Item 3 receives history support from item 0 but no group extra. Item 4 has no history support.

For a concrete illustration, suppose the selected settings use only the standardized $p + \delta$ as Y, and the second estimate is half of a standardized description score. Suppose those description scores after standardization are approximately $(-1.265, -0.632, 0, 0.632, 1.265)$, and use $\kappa = 1$. The final scores for the three candidates are approximately

Candidate	History count f_i	Weight u_i	Final score
2	1	1/2	0.288
3	1	1/2	−0.273
4	0	1	0.632

The top two suggestions are therefore items 4 and 2. Items 0 and 1 were included when standardizing the evidence but are removed from the suggestion list because they were already known. The values chosen here illustrate the calculation; different data and held-out selection examples can lead to different weights and rankings.

12 Choosing suggestions and measuring their quality

The example produced a score for each candidate, but a practical answer needs a short list. After forming that list, we need a separate way to check whether it recovers items that were actually hidden. This final step connects the score to a measurable completion task.

Once every item has a final score $\widehat{\theta}_i$, remove all items already in S. Sort the remaining items by decreasing score and return the first K, or all remaining items if fewer than K are available. When two scores are exactly equal, order the tied items by their item labels, from smaller to larger. This makes the answer repeatable.

For evaluation, assume K does not exceed the number of available candidates. Take a complete bundle and hide some items. Let G be its nonempty set of hidden items, and let R_K be the top-K suggestions from the revealed items. Recall at K is

$$\text{Recall}\,@K = \frac{|R_K \cap G|}{|G|}. \tag{12.1}$$

It measures what fraction of the hidden items were recovered. It does not distinguish a hit at rank 1 from a hit at rank K.

To value earlier hits more, let $h_r = 1$ when the item at rank r is hidden ground truth, and $h_r = 0$ otherwise. The discounted gain and its best possible value are

$$\text{DCG}\,@K = \sum_{r=1}^{K} \frac{h_r}{\log_2(r+1)}, \tag{12.2}$$

$$\text{IDCG}\,@K = \sum_{r=1}^{\min(K,|G|)} \frac{1}{\log_2(r+1)}. \tag{12.3}$$

The normalized discounted gain is $\text{NDCG}\,@K = \text{DCG}\,@K / \text{IDCG}\,@K$. A hit near the top contributes more. Scores for multiple bundles are averaged, giving every evaluated bundle the same weight. Selection examples choose settings; a separate test set measures their performance after those settings are fixed.

13 What the theory does and does not establish

The method combines several interpretable pieces of evidence, and the exposure-based average offers a clear way to rely more on descriptions when history is scarce. The construction alone does not prove that the pieces improve a particular dataset or that the normal-distribution interpretation is accurate. Those questions require comparisons on matched data, splits, inputs, and evaluation rules. Scores and rankings describe the method's preferences under the available evidence; they do not establish causation or guarantee that all recommended items are appropriate for every user.